

What does the Mediterranean diet actually do for health — and how strong is the evidence?

ABSTRACT — The Mediterranean diet is the most extensively studied dietary pattern in medicine. The evidence is strongest for cardiovascular disease: randomised trials in primary prevention and in two secondary-prevention settings show roughly 20–30% fewer major cardiovascular events, and large cohort studies on three continents associate higher adherence with about 10–25% lower mortality, with a dose–response gradient. For type 2 diabetes, trial and observational evidence agree most closely, supported by an objective blood-biomarker measure of adherence. Evidence for other outcomes is weaker: effects on depression, liver fat, weight, and some cancers rest on small or observational studies, and for cognition the cohort studies show inverse associations while the largest dedicated trial found no effect. The evidence base has recognised limitations. Most of it is observational and subject to confounding by other healthy behaviours; associations attenuate outside Mediterranean countries; trial benefits were obtained with intensive dietary support; and the largest primary-prevention trial was retracted and republished after randomisation errors, although the corrected estimates were essentially unchanged. On current evidence, the diet probably prevents cardiovascular events and type 2 diabetes, is associated with modestly lower mortality, and has partly demonstrated mechanisms, while several widely reported benefits remain unconfirmed at trial standard.

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Introduction

The traditional diet of the olive-growing Mediterranean has been proposed as protective against a wide range of chronic diseases. Claimed benefits now span mortality, cardiovascular disease, diabetes, cancer, dementia, depression, liver disease, and [frailty](#), and an [umbrella review](#) rates the evidence robust for some of these outcomes and only suggestive for others [Dinu et al. 2018](#). The pattern is unusual among diets in having hard-endpoint randomised trials, so the observational evidence can be checked against experiment [Estruch et al. 2018](#), [de Lorgeril et al. 1999](#), [Delgado-Lista et al. 2022](#).

The evidence base also has structural weaknesses. Nearly all of the mortality evidence is observational — a Dietary Guidelines systematic review identified 152 [cohort](#)-type studies and a single randomised trial — and the largest primary-

prevention trial was retracted and republished after problems with its randomisation came to light [English et al. 2021](#), [Estruch et al. 2018](#). This review therefore examines the evidence outcome by outcome, from the best-supported claims to the weakest, with contrary evidence considered separately for each: what the diet is, where the claim came from, what trials and cohorts show for each outcome, which mechanisms have been measured, and which limitations qualify the headline estimates.

Definition and measurement

"Mediterranean diet" names a scored pattern, not one menu. Research definitions converge on abundant vegetables, fruit, legumes, and unrefined cereals; olive oil as the principal fat, giving a high ratio of [monounsaturated](#) to saturated fat; regular fish; little red or processed meat; moderate dairy; and moderate wine with meals [Bach et al. 2006](#), [Menotti & Puddu 2025](#). No single food defines the pattern; the score rises with the overall profile.

Adherence is measured with a priori scores rather than by self-identification. The original 0–8/0–9 Trichopoulou scale and its descendants dominate the cohort literature, while trials mostly use the 14-item Mediterranean Diet Adherence Screener (MEDAS) [Trichopoulou et al. 1995](#), [Bach et al. 2006](#). Dozens of score variants now exist, with different components and cut-offs, and this variety complicates every meta-analysis in the field, because a "2-point increment" does not measure the same contrast from one study to the next [Bach et al. 2006](#), [Dinu et al. 2018](#).

Origins of the hypothesis

The health claim originated in ecological data, not experiment. In the [Seven Countries Study](#) — 15 cohorts, 11,579 men, 15 years of mortality follow-up — coronary death rates were lowest in the cohorts where olive oil was the main dietary fat, and the monounsaturated-to-saturated fat ratio was correlated with both all-cause and coronary mortality across cohorts [Keys et al. 1986](#). The association was ecological, across populations, and its authors made no causal claim [Keys et al. 1986](#).

This ecological finding shaped subsequent research. [Ancel Keys'](#) books and the study's cohort contrasts defined the diet as a research object [Menotti & Puddu 2025](#), [Blackburn 2017](#). Later re-analyses of the study's food data produced adherence scores whose ecological correlation with 50-year coronary mortality across cohorts reached -0.91 — still ecological evidence, but consistent, and an indication that the modern scores formalise an old observation rather than an independent discovery [Menotti & Puddu 2025](#).

Cardiovascular outcomes: the trials

Randomised trials in three settings show fewer major cardiovascular events on the

Mediterranean diet. In primary prevention, the PREDIMED trial randomised 7,447 adults at high cardiovascular risk to a Mediterranean diet supplemented with extra-virgin olive oil, the same diet supplemented with nuts, or low-fat advice; over 4.8 years, major cardiovascular events occurred in 3.8% and 3.4% of the two Mediterranean arms against 4.4% of controls, hazard ratios 0.69 (95% CI 0.53–0.91) and 0.72 (0.54–0.95) [Estruch et al. 2018](#).

Secondary prevention was tested first, with a much larger apparent effect. The Lyon Diet Heart Study randomised 605 patients after myocardial infarction and, over 46 months, recorded cardiac death plus nonfatal infarction in 14 patients on the Mediterranean-style diet against 44 on a prudent Western diet, adjusted risk ratios 0.28–0.53 [de Lorgeril et al. 1999](#). An effect of that magnitude has never been replicated, and it is best interpreted as an early outlier rather than the expected yield of the diet [de Lorgeril et al. 1999](#).

A later secondary-prevention trial found an intermediate effect. CORDIOPREV followed 1,002 patients with coronary disease for seven years and found 87 major events on the Mediterranean diet against 111 on a low-fat diet — 28.1 versus 37.7 events per 1,000 person-years, adjusted hazard ratios 0.72–0.75 [Delgado-Lista et al. 2022](#). The benefit was evident in men but not demonstrated in the trial's 175 women, a subgroup too small for a reliable estimate [Delgado-Lista et al. 2022](#).

Pooled across randomised trials, the pattern holds for events but not for deaths: composite cardiovascular events fell (RR 0.63, 95% CI 0.53–0.75) while all-cause mortality did not change (RR 1.00, 0.86–1.15) [Liyanage et al. 2016](#). Table 1 summarises the landmark trials, including two — DIRECT on weight and the MIND trial on cognition — that are discussed in later sections.

TRIAL	SETTING, N	COMPARATOR	MAIN RESULT (95% CI)	REF
Lyon Diet Heart (1999)	Post-MI, 605	Prudent Western diet	Cardiac death + MI: 14 vs 44 events; adjusted RR 0.28–0.53	de Lorgeril et al. 1999
PREDIMED (2018 republication)	Primary prevention, high risk, 7,447	Low-fat advice	MACE HR 0.69 (0.53–0.91) olive oil; 0.72 (0.54–0.95) nuts	Estruch et al. 2018
CORDIOPREV (2022)	Coronary disease, 1,002	Low-fat diet	MACE HR 0.72–0.75; 28.1 vs 37.7 events/1,000 person-years	Delgado-Lista et al. 2022
DIRECT (2008)	Obesity, 322	Low-fat / low-carb	Weight –4.4 kg (Med) vs –2.9 (low-fat) vs –4.7 (low-carb) at 2 y	Shai et al. 2008
MIND diet trial (2023)	Cognition, older adults, 604	Calorie-matched usual diet	Global cognition difference 0.035 SD (–0.022 to 0.092), null	Barnes et al. 2023

The PREDIMED retraction

The 2013 PREDIMED report was retracted and republished in 2018 after randomisation errors were identified. A statistical screen of baseline data across 5,087 trials flagged improbable distributions, which prompted close examination of PREDIMED's randomisation [Carlisle 2017](#). The investigators found protocol deviations — household members enrolled without individual randomisation, and randomisation misapplied at two of the trial's eleven sites — withdrew the 2013 report, and republished the trial with analyses that no longer assumed full randomisation [Estruch et al. 2018](#).

The corrected estimates were essentially unchanged. Omitting the 1,588 participants with known or suspected departures from randomisation left the hazard ratios at 0.69 and 0.72 for the two Mediterranean arms [Estruch et al. 2018](#). The episode exposed conduct failures in the field's central trial and, at the same time, showed the headline result to be robust to their correction.

Two further caveats apply to how the trial is cited. The composite result was driven mainly by stroke — hazard ratio 0.60 (95% CI 0.45–0.80), roughly 24 down to 14 strokes per 1,000 — rather than by infarction or death [Rees et al. 2019](#). And PREDIMED showed no effect on total mortality (HR 1.0, 0.81–1.24), an outcome for which it was not powered; the trial demonstrates fewer events, not lower mortality [Guasch-Ferré et al. 2017](#), [Rees et al. 2019](#).

What the syntheses conclude

Independent syntheses support the cardiovascular benefit at moderate certainty. A [network meta-analysis](#) of 40 trials with 35,548 patients

rated Mediterranean dietary programmes superior to minimal intervention with moderate [GRADE](#) certainty: all-cause mortality OR 0.72 (95% CI 0.56–0.92), cardiovascular mortality 0.55 (0.39–0.78), stroke 0.65 (0.46–0.93), and nonfatal infarction 0.48 (0.36–0.65), amounting to about 17 fewer deaths per 1,000 over five years at intermediate risk [Karam et al. 2023](#). An umbrella review of trial meta-analyses put cardiovascular risk ratios in the same range, 0.55–0.64 [Chareonrungrueangchai et al. 2020](#).

The choice of comparator conditions these results. The same network meta-analysis found no convincing advantage of Mediterranean over low-fat programmes for mortality or infarction — the diet outperforms minimal intervention, but not clearly the main structured alternative [Karam et al. 2023](#). The Cochrane review reaches a more conservative conclusion: across 30 randomised trials and 12,461 participants it finds that "some uncertainty" remains, with risk-factor effects that are small — systolic blood pressure –3.0 mmHg (95% CI –3.5 to –2.5), total cholesterol –0.16 mmol/L (–0.32 to 0.00) — and evidence quality low to moderate throughout [Rees et al. 2019](#). The overall picture is a benefit of moderate certainty against minimal intervention, no demonstrated superiority over other structured diets, and small effects on classical risk factors.

Mortality in the cohorts

Large cohort studies consistently associate higher adherence with lower mortality, with a dose-response gradient. In the founding cohort of 22,043 Greek adults, each 2-point increment on the adherence score predicted 25% lower total mortality (HR 0.75, 95% CI 0.64–0.87), and across 74,607 elderly Europeans the same increment predicted an 8% reduction (3–12%) [Trichopoulou et al. 2003](#), [Trichopoulou et al.](#)

2005. The finding replicates outside Europe: in the [Nurses' Health Study](#) and [Health Professionals cohorts](#), top-quintile Alternate Mediterranean scores predicted mortality HR 0.82 (0.79–0.84) over up to 36 years and cardiovascular disease HR 0.83 (0.79–0.86), and in 25,315 US women followed up to 25 years the top-adherence hazard ratio was 0.77 (0.70–0.84) [Shan et al. 2023](#), [Shan et al. 2020](#), [Ahmad et al. 2024](#).

The pooled effect is smaller than the early Greek estimates. Meta-analyses of cohorts converge near HR 0.90 per 2 points (0.89–0.91; 29 studies, 1.68 million people) and 0.79 (0.77–0.81) for highest-versus-lowest adherence, and the newest synthesis of 54 cohorts and 1.83 million people grades the evidence moderate at RR 0.96 (0.95–0.97) per single point [Sofi et al. 2008](#), [Soltani et al. 2019](#), [Eleftheriou et al. 2018](#), [Nucci et al. 2026](#). The attenuation from HR 0.75 to about 0.90 per 2 points as the literature accumulated indicates a consistent association whose size the founding cohort overestimated.

The association also holds in higher-risk groups and is not attributable to any single food. It is present in older adults (all-cause mortality RR 0.77, 0.70–0.83) and in people with diabetes (2-point HR 0.63; pooled RR 0.93, 0.90–0.97) [Furbatto et al. 2024](#), [Bonaccio et al. 2016](#), [Muscogiuri et al. 2026](#). Component analyses attribute most of the mortality association to moderate alcohol (~24%), low meat (~17%), and high vegetable intake (~16%), with cereals, dairy, and fish contributing little — consistent with an effect of the overall pattern rather than of any single component [Trichopoulou et al. 2009](#).

Type 2 diabetes

Trial and cohort evidence agree most closely for diabetes prevention. Within PREDIMED, the pooled Mediterranean arms reduced incident diabetes (HR 0.70, 95% CI 0.54–0.92), and yearly-measured adherence showed a gradient — HR 0.80 (0.70–0.92) per 2 points and 0.46 (0.25–0.83) for highest versus lowest adherence [Bloomfield et al. 2016](#), [Martínez-González et al. 2023](#). Adding energy restriction and exercise increases the effect: PREDIMED-Plus, with 4,746 adults over six years, reduced diabetes incidence a further 31% (95% CI 18–41%) relative to an

unrestricted Mediterranean diet, an absolute risk of 9.5% versus 12.0% [Ruiz-Canela et al. 2025](#).

The diet also improves established diabetes. Meta-analyses of randomised trials show better glycaemic control, with [HbA1c](#) reductions of 0.30–0.47 percentage points and BMI -0.83 (95% CI -1.40 to -0.26), and long-term trials show a 49% higher probability of remission from the metabolic syndrome [Esposito et al. 2015](#), [Wu et al. 2025](#).

Two independent checks indicate that this signal is not an artefact of self-report. A blood biomarker score of Mediterranean eating — carotenoids plus fatty acids, validated in a feeding trial — predicted 29% lower diabetes incidence per standard deviation (HR 0.71, 0.65–0.77) in 22,202 EPIC-InterAct participants, a larger effect than self-reported scores suggest [Sobiecki et al. 2023](#). And in the non-Mediterranean UK Biobank, a Mediterranean lifestyle score predicted 30% lower diabetes incidence (top-quartile HR 0.70, 0.62–0.79), showing that the association transfers across settings [Maroto-Rodriguez et al. 2023](#).

Risk factors, weight, and body composition

Cardiometabolic risk factors improve consistently but modestly on the diet. Meta-analyses of trials and cohorts show small shifts across the metabolic syndrome's components — waist circumference -0.42 cm, triglycerides -6.1 mg/dl, systolic blood pressure -2.4 mmHg, glucose -3.9 mg/dl — and a pooled metabolic syndrome risk ratio around 0.81 [Kastorini et al. 2011](#), [Godos et al. 2017](#), [Papadaki et al. 2020](#). The individual effects are small; they are notable for occurring together and persisting over time.

For weight loss, the diet is comparable to other structured diets, with relatively durable effects. In the 2-year DIRECT trial it outperformed low fat (-4.4 vs -2.9 kg) and matched low carbohydrate (-4.7 kg), and a network meta-analysis found it the only named diet whose cardiovascular risk-factor benefits persisted at 12 months [Shai et al. 2008](#), [Ge et al. 2020](#). Body-composition trials are consistent with this: an energy-restricted Mediterranean diet plus activity reduced total and visceral fat and attenuated lean-mass loss over three years, and a polyphenol-enriched "green" variant more than doubled visceral-fat loss (-14.1% vs -6.0%) [Konieczna et al. 2023](#), [Zelicha et al. 2022](#).

Contrary evidence exists at the risk-factor level. A 2024 meta-analysis of 18 trials of Mediterranean diet enriched with olive oil found no consistent benefit on any cardiometabolic marker except triglycerides (-2.4 mg/dl, 95% CI -4.5 to -0.3), and a GRADE-based review in overweight adults rated most of the risk-factor evidence very low quality [Keshani et al. 2024](#), [Hernandez et al. 2025](#). Supplementing one component is not equivalent to changing the pattern, and the risk-factor evidence is weaker than the evidence on clinical events.

Cancer

Observational associations with cancer are broad but mostly weak. Cohort meta-analyses associate the highest adherence with lower cancer mortality (RR 0.86–0.87) and lower colorectal cancer incidence (0.82–0.83), with a weaker association for breast cancer (0.94, 95% CI 0.90–0.97) [Morze et al. 2021](#), [Schwingshackl et al. 2017](#). An umbrella review rates the evidence robust only for overall cancer incidence and mortality; for most site-specific cancers it is suggestive or weak, and for several sites null [Dinu et al. 2018](#), [Morze et al. 2021](#).

Trial evidence is limited to a single secondary outcome. PREDIMED's olive-oil arm recorded fewer invasive breast cancers (HR 0.32, 95% CI 0.13–0.79), but with only 35 cases in total the finding requires confirmation before it carries weight [Toledo et al. 2015](#). Olive oil itself shows the same asymmetry: at $+25$ g/day it is associated with lower cardiovascular disease, diabetes, and mortality across 1.29 million participants, but not with lower cancer risk (RR 0.94, 0.86–1.03) [Martínez-González et al. 2022](#). For cancer, the evidence supports an association, with no demonstrated causal protection for any specific site.

Cognition

Trial and cohort evidence diverge most clearly for cognition. The cohorts and their meta-analyses are consistently positive: cognitive impairment HR 0.82 (0.75–0.89), dementia 0.89 (0.83–0.95), and Alzheimer's disease 0.70 (0.60–0.82)

[Fekete et al. 2025](#), [Fu et al. 2022](#). In 60,298 UK Biobank participants, top-tertile adherence predicted 23% lower dementia risk (HR 0.77, 0.65–0.91) independent of polygenic risk, and the related MIND diet shows a pooled dementia hazard ratio of 0.83 (0.76–0.90) across cohorts [Shannon et al. 2023](#), [Chen et al. 2023](#).

Early trial evidence appeared concordant. PREDIMED's Navarra sub-study found better MMSE scores after 6.5 years on the olive-oil diet ($+0.62$ points, 95% CI 0.18–1.05, versus control), and pooled randomised trials show an episodic-memory benefit of SMD 0.20 (0.09–0.30) [Martínez-Lapiscina et al. 2013](#), [Fu et al. 2022](#).

The largest dedicated trial, however, was null. The 3-year MIND trial randomised 604 older adults to the MIND diet or a calorie-matched usual diet, both with mild calorie restriction, and found a global-cognition difference of 0.035 SD (95% CI -0.022 to 0.092 , $p=0.23$) and no differences on MRI; both groups improved and both lost about 5 kg [Barnes et al. 2023](#). Plausible explanations include an unusually active, weight-losing control arm, a well-nourished volunteer population, and residual confounding in the cohorts [Barnes et al. 2023](#), [Maki et al. 2014](#). The trial cannot rule out effects over longer durations or in higher-risk populations, but it constrains what can be claimed for three years of dietary change in healthy older volunteers [Barnes et al. 2023](#).

Depression

Small randomised trials report the largest effects for depression, but the pooled evidence is low-certainty. Observationally, higher adherence is associated with 33% lower incident depression (OR 0.67, 95% CI 0.55–0.82) [Lassale et al. 2019](#). Pooling five randomised trials with 1,507 participants, depressive symptoms fell by SMD -0.53 (95% CI -0.90 to -0.16) — a moderate effect by convention, and larger than most drug-placebo differences in mild depression [Bizzozero-Peroni et al. 2025](#). Table 2 shows the three trials that drive the pooled estimate.

TRIAL	N, DURATION	CONTROL	RESULT	REF
SMILES (2017)	67 adults, major depression, 12 wk	Social support	<u>MADRS</u> −7.1 points vs control; re-mission 32% vs 8%; <u>NNT</u> 4.1	<u>Jacka et al. 2017</u>
HELFIMED (2019)	152 adults, self-reported depression, 3–6 mo	Social groups	Greater depression reduction at 3 mo, sustained at 6 mo	<u>Parletta et al. 2019</u>
AMMEND (2022)	72 young men, moderate–severe depression, 12 wk	Befriending	BDI-II improved 14.4 points more (95% CI 11.4–17.4)	<u>Bayes et al. 2022</u>

Several features limit confidence in the pooled estimate. Heterogeneity across the trials is high ($I^2 = 87\%$), the prediction interval spans zero (−1.86 to 0.81), and the certainty of evidence is rated low Bizzozero-Peroni et al. 2025. All three positive trials are small, unblinded to participants, and use comparators — befriending, social groups — that may underperform genuine usual care, which would inflate the apparent benefit of the diet arm Bizzozero-Peroni et al. 2025. The direction of effect is consistent; its size remains uncertain.

Liver, frailty, and joints

For fatty liver disease, most small trials show benefit, with one dissenting result. A meta-analysis of six small randomised trials shows reduced steatosis (fatty-liver index SMD −1.06, 95% CI −1.95 to −0.17) and improved insulin resistance, and a crossover trial confirmed steatosis reduction even without weight loss Kawaguchi et al. 2021, Ryan et al. 2013. The 18-month DIRECT-PLUS trial reported larger effects, nearly doubling intrahepatic-fat loss with a green-Mediterranean variant (−38.9% versus −12.2% proportional loss on guideline care) Yaskolka Meir et al. 2021. Against this, one 12-week trial found no advantage over a low-fat diet, and clinical reviews continue to place calorie restriction, not diet composition, at the centre of NAFLD management George et al. 2022, Haigh et al. 2022.

Evidence on frailty is observational, and on joint disease sparse. Higher adherence predicts roughly half the risk of incident frailty in Spanish cohorts (odds ratios 0.48 and 0.38 for top adherence) and lower frailty plus 8-year mortality in US NHANES participants León-Muñoz et al. 2014, Maroto-Rodriguez et al. 2022, Jayanama et al. 2021. In rheumatoid arthritis, a systematic review finds improvement in some disease-activity measures, smaller and less consistent than the effects of omega-3 supplementation Philippou et al. 2021. These findings are preliminary.

Mechanisms

Several proposed mechanisms have been measured directly, some within the hard-endpoint trials themselves. In CORDIOPREV, one year of the Mediterranean diet produced better endothelial function than low fat — flow-mediated dilation 3.83% (95% CI 2.91–4.23) versus 1.16% (0.80–1.98) — and by 5–7 years carotid intima-media thickness had regressed (−0.027 mm) while the low-fat arm's had not Yubero-Serrano et al. 2020, Jimenez-Torres et al. 2021. The same trial population thus shows changes in the artery wall alongside changes in event rates.

Inflammation is a second measured pathway. Top-quintile Mediterranean scores predict roughly 24% lower C-reactive protein (CRP), and in the Women's Health Study, inflammatory and small-molecule metabolite biomarkers were the largest single mediators of the diet–mortality association Fung et al. 2005, Ahmad et al. 2024.

More recent mechanistic work concerns the gut microbiome and markers of aging. One-year interventions across five countries shifted the gut microbiota toward short-chain-fatty-acid producers, tracking lower CRP and IL-17 and better frailty markers, and shorter trials confirm microbiome and bile-acid shifts alongside improved insulin sensitivity Ghosh et al. 2020, Meslier et al. 2020, Muralidharan et al. 2021, Rinott et al. 2022. The polyphenol-rich green-Mediterranean variant has, in addition, attenuated epigenetic-clock aging and age-related brain atrophy in participants over 50 Yaskolka Meir et al. 2023, Kaplan et al. 2022. Mechanistic findings do not substitute for outcome data, but they reduce the plausibility that the associations are wholly attributable to confounding.

Limitations of the evidence

The principal limitation is that almost all of the mortality evidence is observational. The Dietary Guidelines review identified 152 cohort-type studies against one randomised trial, in a literature where adherent eaters also exercise more,

smoke less, and have higher socioeconomic status, and no adjustment removes all of that confounding [English et al. 2021](#), [Maki et al. 2014](#), [Ioannidis 2018](#). Measurement error compounds the confounding: self-reported food-frequency instruments are noisy, and the dozens of non-interchangeable adherence scores in circulation make effects hard to compare across studies [Bach et al. 2006](#), [Ioannidis 2018](#).

Generalisability is a second, quantified limitation. The pooled mortality association is HR 0.82 per 2 points inside Mediterranean countries but 0.92 elsewhere — the effect attenuates with distance from the Mediterranean, whether because of food quality, dose, or culture [Soltani et al. 2019](#). Trial effects, meanwhile, depend on intensive support — free foods, dietitian counselling, quarterly group sessions — so unsupported real-world adoption will likely deliver less, and some trials are part-funded by olive-oil sector foundations [Rees et al. 2019](#), [Karam et al. 2023](#), [Delgado-Lista et al. 2022](#).

The umbrella-review assessment summarises the balance: robust evidence for mortality, cardiovascular disease, overall cancer incidence, neurodegenerative disease, and diabetes; suggestive or weak evidence for most other outcomes [Dinu et al. 2018](#). These caveats do not overturn the cardiovascular trials; they caution against extrapolating cohort-derived estimates to unsupported adoption in non-Mediterranean settings.

Open questions

Several ongoing studies address these gaps. Hard-endpoint results from PREDIMED-Plus — 6,874 participants in an 8-year cardiovascular follow-up of energy-restricted Mediterranean diet plus exercise — will test whether adding weight loss amplifies the PREDIMED effect [Konieczna et al. 2023](#), [Ruiz-Canela et al. 2025](#). Objective adherence biomarkers, metabolomic and carotenoid/fatty-acid scores among them, can reduce the reporting bias in cohort estimates [Sobiecki et al. 2023](#), [Martínez-González et al. 2023](#). Polyphenol-dense green-Mediterranean variants are testing whether the pattern's effects can be enhanced [Yaskolka Meir et al. 2021](#), [Zelicha et al. 2022](#).

For cognition and depression, the priority is trial quality rather than trial number: longer and larger studies in higher-risk or non-Mediterranean populations, with credible control arms — the

MIND trial's active, weight-losing comparator illustrates how a strong control condition can eliminate an expected effect [Barnes et al. 2023](#), [Bizzozero-Peroni et al. 2025](#), [Wu et al. 2025](#).

Conclusion

The Mediterranean diet's evidence base is unusual in nutrition science: hard-endpoint randomised trials exist, they broadly agree with a large observational literature, and the central trial's estimates were essentially unchanged after retraction and republication. Taken together, the diet probably reduces major cardiovascular events by a fifth to a third against minimal intervention, probably prevents type 2 diabetes, and is consistently associated with roughly 10–25% lower mortality at high adherence — a real but modest association that attenuates outside the Mediterranean and has not been demonstrated in a trial powered for mortality.

Confidence is lower for the remaining outcomes. Weight and risk-factor effects are small and partly low-certainty; the cancer association is broad but causally undemonstrated; liver, frailty, and joint findings are preliminary; depression shows a consistent direction of effect with an uncertain size; and cognition shows a direct conflict between cohort associations and the largest trial. The studies most likely to resolve these questions are identifiable: PREDIMED-Plus for hard endpoints, biomarker-measured adherence to discipline the cohort estimates, and adequately controlled trials for the outcomes where the evidence is weakest. On current evidence, the Mediterranean diet is one of few dietary patterns with trial-level evidence of benefit — at modest effect sizes — for cardiovascular events and diabetes, with consistent observational support for lower mortality.

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